

View results

Respondent

42

Anonymous

55:36

Time to complete

A. General information about the respondent

1. * Country

Philippines



2. * Name of organization(s) or entity(s) responsible for the preparation of the report

Department of Science and Technology

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

<https://www.dost.gov.ph/>

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

The Department of Science and Technology provides central direction, leadership and coordination of scientific and technological efforts and ensure that the results therefrom are geared and utilized in areas of maximum economic and social benefits for the people.

7. * Full name of officially designated contact person(s) who completed this survey

Leah J. Buendia

8. * Position of officially designated contact person(s) who completed this survey

Undersecretary for Research and Development

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

Leah J. Buendia

11. Brief description of the consultation process established for the preparation of the report

The other organizations were consulted through self-administered completion of survey questions found in this report. However, due to limited time from the day of designation as the focal point for this report, responses received have been limited.

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

Philippine Council for Agriculture, Aquatic and Natural Resources Research and Development (DOST-PCAARRD); Philippine Council for Industry, Energy, and Emerging Technology Research and Development (DOST-PCIEERD); Philippine Council for Health Research and Development (DOST-PCHRD)

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey (Please use commas to separate multiple links, as: link A, link B)

<https://www.pcaarrd.dost.gov.ph/>, <https://pcieerd.dost.gov.ph/>, <https://www.pchrd.dost.gov.ph/>

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey



Public



Private



Civil Society



Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN
SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?



Yes



No

17. 1.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

The promotion of the 2021 UNESCO Recommendation on Open Science remains an area for future progress in the Philippines. Engagement with international frameworks, such as this Recommendation could further guide the Philippines in formalizing its commitment to Open Science through comprehensive national policies and implementation plans.

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

☐ Yes

☒ No

19. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) 16.f, 16.g, 16.h)

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

☐ Yes

☒ No

20. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

☒ Yes

☐ No

21. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

• DOST Grants-In-Aid (GIA) Program (<https://dpmis.dost.gov.ph/index.php/transparency/about-us>;
https://www.dost.gov.ph/phocadownload/Downloads/Resources/Guidelines/2020/Administrative_Order_No_%20011_series_of_2020_Revised_DOST-GIA_Guidelines.pdf) • Good Governance through Data Science and Decision Support Systems (GODDESS) Program that develops technologies or systems using available data from institutions or collaborating partners.
<https://pcieerd.dost.gov.ph/supported-programs-pprojects/strategic-thrusts-and-programs/239-human-resource-development-2#xi-good-governance-through-data-science-and-decision-support-system-goddess> • Collaborative Research and Development to Leverage Philippine Economy (CRADLE) Program that focuses on the improvement government-industry-academe collaboration in addressing pressing problems of the country
(https://www.dost.gov.ph/phocadownload/Downloads/Resources/SCIENCE_FOR_CHANGE_PROGRAM_S4CP/CRADLE_brochure.pdf)

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

22. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

☒ Yes

☐ No

23. 2.1 cont. **If yes**

- please share a machine-readable PDF file of the policy using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

Science, technology, and innovation is mostly incorporated in the national plans of the Philippines: (i) Philippine Development Plan (PDP) 2023-2028 (<https://pdp.neda.gov.ph/philippine-development-plan-2023-2028/>); and (ii) AmBisyon Natin 2040 (<https://2040.neda.gov.ph/about-ambisyon-natin-2040/>). The Department of Science and Technology (DOST) serves as the lead agency responsible for implementing and coordinating STI policies, programs, and initiatives in the country. It plays a crucial role in shaping the national STI agenda through key frameworks such as: 1. Harmonized National Research and Development Agenda (HNRDA) 2022-2028 – Aligns national R&D priorities across agriculture, industry, health, disaster resilience, and emerging technologies to ensure strategic and effective resource allocation. (<https://www.dost.gov.ph/knowledge-resources/downloads/file/5669-harmonized-national-r-d-agenda-2022-2028.html>); 2. DOST Pagtanaw 2025 - inter-disciplinal and trans-disciplinal project on Philippine-focused STI Foresight and Strategic Plan that will impact on the aspirations of the Filipino people by 2050. (https://www.dost.gov.ph/phocadownload/Downloads/Resources/Quick_Links/HNRDA_2022-2028.pdf); and 3. Philippine Innovation Act (RA 11293) – Establishes a national innovation framework that promotes STI-led growth, enhances competitiveness, and supports research commercialization. This is a joint administrative order by key national government agencies in the Philippines. (https://www.dost.gov.ph/phocadownload/Downloads/Resources/RA/IRR_of_RA-11293.pdf)

24. 2.1 cont. **If yes:**

- does it include any of the following open science elements:

- ☒ open access to scientific knowledge
- ☒ open science infrastructure
- ☒ open engagement of societal actors
- ☒ open dialogue with other knowledge systems

25. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ☒ Yes
- ☐ Partly
- ☐ No

26. 2.2 cont. **If yes or partly**, please share a machine-readable PDF file(s) of the policy(ies) using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

1. The Technology Transfer Act of 2009 (Republic Act No. 10055) is considered a key policy instrument for promoting open science in the Philippines because it facilitates the dissemination and commercialization of research outputs generated by publicly funded research and development (R&D) activities. The law encourages transparency, collaboration, and sharing of research results by mandating that intellectual property rights from government-funded research be owned by the research institutions rather than the state. This framework promotes open access to scientific innovations while allowing researchers and institutions to engage with industry and other stakeholders for commercialization purposes. The Act also ensures that the benefits of publicly funded research are accessible to a wider audience, aligning with the principles of open science by fostering knowledge sharing, innovation, and broader societal impact. (<https://www.dost.gov.ph/knowledge-resources/2014-04-27-01-59-53/republic-acts/file/288-republic-act-no-10055-philippine-technology-transfer-act-of-2009.html>) 2. The Philippine Innovation Act (Republic Act No. 11293) promotes open science through collaborative research, sharing of scientific data and knowledge, and the encouragement of partnerships between the academe, government, and industry. The law aims to strengthen the country's innovation governance system through fostering a culture of innovation across all sectors, enhancing productivity, and improving competitiveness. The National Innovation Council (NIC), chaired by the President of the Philippines with the National Economic and Development Authority (NEDA) as its secretariat, is the primary authority responsible for implementing the law. The NIC oversees the formulation and implementation of the National Innovation Agenda and Strategy Document (NIASD), which outlines the country's innovation goals and priorities. 3. The Philippine Space Agency (PhilSA) (Republic Act No. 11363) promotes open science through its initiatives focused on transparency, accessibility, and collaboration in space research and data sharing. PhilSA actively works to make space science and satellite data accessible to government agencies, academic institutions, and the public to support research, policy-making, and development initiatives, especially in areas such as disaster risk reduction, environmental monitoring, and agricultural productivity. (<https://philsa.gov.ph/philippine-space-act/>) 4. The Philippine Ecosystem and Natural Capital Accounting System (PENCAS) Act (Republic Act No. 11995) also demonstrates elements of open science by promoting transparency, accessibility, and the collaborative use of environmental and economic data. PENCAS institutionalizes the systematic collection, analysis, and dissemination of natural capital accounts. These accounts help inform evidence-based policymaking, sustainable development, and resource management by making data accessible to researchers, policymakers, and the public. The Philippine Statistics Authority (PSA) serves as the lead authority in implementing PENCAS. <https://faolex.fao.org/docs/pdf/phi228431.pdf>; 5. The Philippine National Health Research System (PNHRS) Act of 2013 (Republic Act 10532) is a law that established the PNHRS as a framework anchored on the principles of Essential National Health Research on inclusiveness, participation, quality, equity, efficiency and effectiveness, which connect to, and converge with, the wider health, economic, political, educational and S&T systems of the country. Its aims include engaging in national and international partnerships and networks

for health research development, and ensuring sustainability of resources for health research, among others. <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/96663>

27. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).



Yes



Under development



No

28. 2.3 cont. **If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

1. The Technology Transfer Act of 2009 (Republic Act No. 10055) is considered a key policy instrument for promoting open science in the Philippines because it facilitates the dissemination and commercialization of research outputs generated by publicly funded research and development (R&D) activities. The law encourages transparency, collaboration, and sharing of research results by mandating that intellectual property rights from government-funded research be owned by the research institutions rather than the state. This framework promotes open access to scientific innovations while allowing researchers and institutions to engage with industry and other stakeholders for commercialization purposes. The Act also ensures that the benefits of publicly funded research are accessible to a wider audience, aligning with the principles of open science by fostering knowledge sharing, innovation, and broader societal impact. The DOST serves as the primary authority responsible for overseeing the implementation of this act, ensuring compliance, and facilitating partnerships between research institutions and industries. The law does not specify a fixed allocated budget, nonetheless, funding for its implementation is usually sourced from R&D project budgets of agencies under the DOST and supplemented by institutional funds from state, universities, and colleges (SUCs) and research and development institutions (RDIs). Other agencies such as the Department of Trade and Industry (DTI) and the Intellectual Property Office (IPO) also play a key role in the implementation of the act. (<https://www.dost.gov.ph/knowledge-resources/2014-04-27-01-59-53/republic-acts/file/288-republic-act-no-10055-philippine-technology-transfer-act-of-2009.html>); 2. The Philippine Innovation Act (Republic Act No. 11293) promotes open science through collaborative research, sharing of scientific data and knowledge, and the encouragement of partnerships between the academe, government, and industry. The law aims to strengthen the country's innovation governance system through fostering a culture of innovation across all sectors, enhancing productivity, and improving competitiveness. The National Innovation Council (NIC), chaired by the President of the Philippines with the National Economic and Development Authority (NEDA) as its secretariat, is the primary authority responsible for implementing the law. The NIC oversees the formulation and implementation of the National Innovation Agenda and Strategy Document (NIASD), which outlines the country's innovation goals and priorities. (https://www.dost.gov.ph/phocadownload/Downloads/Resources/RA/IRR_of_RA-11293.pdf); 3. Harmonized National Research and Development Agenda (HNRDA) 2022-2028 – Aligns national R&D priorities across agriculture, industry, health, disaster resilience, and emerging technologies to ensure strategic and effective resource allocation. (<https://www.dost.gov.ph/knowledge-resources/downloads/file/5669-harmonized-national-r-d-agenda-2022-2028.html>); 4. DOST Grants-In-Aid (GIA) Program aims to harness the country's scientific and technological capabilities to spur and attain sustainable economic growth and development. Through the funding of relevant science and technology (S&T) undertakings, the GIA program aims to contribute to productivity improvement and quality of life of Filipinos by generating and promoting appropriate technologies. Likewise, it seeks to strengthen the participation of various S&T sectors particularly in research and development (R&D), promotion, technology transfer and utilization, human resources development, information dissemination, advocacy, and linkages. (<https://dpmis.dost.gov.ph/index.php/transparency/about-us>; https://www.dost.gov.ph/phocadownload/Downloads/Resources/Guidelines/2020/Administrative_Order_No_%20011_series_of_2020_Revised_DOST-GIA_Guidelines.pdf)

29. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)

- ☐ Yes
- ☐ Under development
- ☒ No

30. 2.4 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

The development of a dedicated national open science strategy or roadmap remains an area for future progress in the Philippines. Engagement with international frameworks, such as the UNESCO Recommendation on Open Science, could further guide the Philippines in formalizing its commitment to open science through comprehensive national policies and implementation plans.

31. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 17.e, 17.g)

- ☒ Yes
- ☐ Under development
- ☐ No

32. 2.5 cont. **If yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

To view the elements of open science,

see: https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/_page=9

Yes, the Philippines has various policies and strategies at the institutional level that promote open science in line with the 2021 UNESCO Recommendation on Open Science. 1. DOST-PCAARRD eLibrary (<https://elibrary.pcaarrd.dost.gov.ph/home>) offers an open-access digital platform that provides researchers with free access to publications, theses, and technical reports in agriculture, aquatic, and natural resources (AANR). DOST-PCAARRD Advanced Learning Management System (PALMS), likewise, is an open-access knowledge-sharing platform that provides training modules, webinars, and capacity-building courses for researchers, students, and policymakers. 2. Established in 2014 at the Advanced Science and Technology Institute (ASTI) of DOST, the Computing and Archiving Research Environment (COARE) is a High-Performance Computing and Cloud facility that allows free access of its services to students, researchers, and data analysts. (<https://asti.dost.gov.ph/projects/coare/>) 3. There are various strategies that promote open science such as the National Science, Technology, and Innovation Week which highlights the significant contributions of science and technology to national development and has become a platform for heralding S&T advocacy in the country. NSTW includes activities for various fields like basic and applied research, technology transfer, technology commercialization and S&T promotion, scientific meetings, technical and investment forums, technology demonstrations, and technical tours. (<http://nstw.dost.gov.ph/>). There are also some activities outside the ministries: • University of the Philippines Open University's research conversation series on open science and free courses: <https://www.upou.edu.ph/announcement/research-conversations-open-science-series/>; • University of Santo Tomas (UST) promotes open science at the UST Science Week celebration: <https://www.ust.edu.ph/open-science-initiatives-highlighted-in-annual-ust-science-week-celebration/> • University of the Philippines Manila's integration of research integrity and ethics.

33. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? (ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.



Yes



Under development



No

34. 2.6 cont. **If yes or under development**, please provide details and links as relevant.

• DOST Grants-In-Aid (GIA) Program (<https://dpmis.dost.gov.ph/index.php/transparency/about-us>;
https://www.dost.gov.ph/phocadownload/Downloads/Resources/Guidelines/2020/Administrative_Order_No_%20011_series_of_2020_Revised_DOST-GIA_Guidelines.pdf) • Startup Grant Fund Program: https://www.pchrd.dost.gov.ph/news_and_updates/pchrd-startup-research-grant-program/; <https://www.pcaarrd.dost.gov.ph/index.php/quick-information-dispatch-qid-articles/dost-pcaarrd-welcomes-the-3rd-batch-of-startup-grant-fund-beneficiaries-and-launches-lipad-phase-2>; <https://pcieerd.dost.gov.ph/e-forms/call-for-proposals/41-work-with-us/521-2023-call-for-proposals-for-the-pcieerd-startup-grant-fund-program-cycle-2> • Science for Change Program (S4CP) • Technology Innovation for Commercialization (TECHNiCOM)
<http://www.tapi.dost.gov.ph/call-for-proposals/technicom>; <http://www.tapi.dost.gov.ph/579-together-to-get-there-with-dost-tapi-2025-dost-tapi-empowers-local-inventors-and-innovators-with-funding-programs>

35. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

☒ Yes

☐ No

36. 2.7 cont. **If yes**, please provide details and links as relevant.

The DOST-GIA Program as indicated in the previous questions has integrated open science practices into its existing research funding mechanisms, ensuring that publicly funded research contributes to open-access knowledge-sharing, data dissemination, and collaboration. Researchers must share their findings through open-access platforms, conferences, and digital repositories. The DOST GIA-funded projects are also required to submit knowledge products such as policy briefs, technical reports, and datasets to be made publicly accessible. The agencies also encourage publication in open-access journals to maximize research visibility and impact. (<https://dpmis.dost.gov.ph/index.php/transparency/about-us>; https://www.dost.gov.ph/phocadownload/Downloads/Resources/Guidelines/2020/Administrative_Order_No_%20011_series_of_2020_Revised_DOST-GIA_Guidelines.pdf)

37. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

☒ Yes

☐ No

38. 2.8 cont. If yes, please provide details and links as relevant.

The Philippines is actively working to establish equitable public-private partnerships (PPPs) in open science, aligning with the 2021 UNESCO Recommendation on Open Science. While formalized initiatives are still in development, efforts are focused on integrating open science principles into government-funded research and fostering collaboration with private sector entities, particularly in technology, data, and AI. Other efforts that foster equitable public-private partnerships on open science in line with the 2021 Recommendation include its (1) DOST Grants-in-Aid Program where HEIs and non-profit organizations are supported to implement their R&D activities and (2) the conduct of the research approved under the Renewable Energy Science, Technology & Innovation (RESTI) Program with the Department of Energy.
(<https://dpmis.dost.gov.ph/index.php/transparency/about-us>;
https://www.dost.gov.ph/phocadownload/Downloads/Resources/Guidelines/2020/Administrative_Order_No_%20011_series_of_2020_Revised_DOST-GIA_Guidelines.pdf)

39. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

- ☒ Yes
- ☐ Under development
- ☐ No

40. 2.9 cont. If yes or under development, please provide details and links as relevant.

Yes, DOST has established a monitoring and evaluation (M&E) protocol that emphasizes the importance of defining clear expectations and providing means to achieve them, thereby optimizing R&D results.
(https://www.dost.gov.ph/phocadownload/Downloads/Resources/Issuances/Administrative_Orders/2019/DOST-AO-014-2019.pdf)

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

41. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

The Philippines has 0.32% GDP to R&D expenditures in 2018 as per World Bank Data publication on the same year and WIPO on the Global Innovation Index Report 2024 - <https://data.worldbank.org/indicator/GB.XPD.RSDV.GD.ZS>; <https://www.wipo.int/edocs/gii-ranking/2024/ph.pdf>

42. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

73.1% in 2023; <https://businessmirror.com.ph/2023/05/09/make-customers-feel-they-are-fairly-treated/>; <https://pidswebs.pids.gov.ph/CDN/PUBLICATIONS/pidspn2107.pdf>; <https://pidswebs.pids.gov.ph/CDN/PUBLICATIONS/pidspn2107.pdf>

43. 3.3 Are there national research and education networks (NRENs)[i] active in the country? (ref.: (iii) 18.c)

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

☒ Yes

☐ No

44. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

Philippine Research, Education, and Government Information Network (PREGINET) is the only REN in the Philippines, which interconnects and catalyzes research among academic, government, and research institutions. PREGINET has links to several overseas RENs including the Asia-Pacific Advanced Network (APAN), Asian Internet Interconnection Initiatives (AI3), and Trans-Eurasia Information Network 3 (TEIN-CC). Compared with commercial ISPs, PREGINET provides value to academic, government, and research institutions in four (4) primary areas: (i) direct connectivity to international RENs; (ii) enhanced communication through its value-added services which include videoconferencing, video streaming, and Voice over IP (VoIP); (iii) access to local and foreign content and applications via conduct of Research and Development (R&D); and (iv) facilitating formation of user-communities in areas of agriculture, bioinformatics, disaster mitigation, distance education, network technologies, telemedicine, among others.
(<https://asti.dost.gov.ph/projects/preginet/#>)

45. 3.4 Are there national or institutional open science infrastructures[i] in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.



Yes



Under development



No

46. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- ☒ federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- ☐ community managed infrastructures, protocols and standards, for example those that support biodiversity and engagement with society
- ☒ platforms for exchanges and co-creation of knowledge between scientists and society
- ☐ community-based monitoring and information systems to complement national, regional and global data and information systems

47. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

The Philippines has the following programs/projects categorized specifically under (i) the platforms for exchanges and co-creation of knowledge between scientists and society and (ii) information technology infrastructure for open science, including high-performance computing, cloud computing and data storage based on the type of infrastructure defined by UNESCO 2021 Recommendation:

- The DOST Technology Hub (TechHub) provides a curated collection of the department's scientific and technological innovations and services. This includes processes on adoption, transfer, and commercialization. It is also a testament to DOST's commitment of bringing science closer to the people. The following DOST priorities are aligned with the President's socioeconomic agenda such as follows: Livelihood, Food, Health, Water & Environment, Energy, Transportation, Climate and Disaster Resiliency, and Digital Transformation. <https://techub.dost.gov.ph/>
- Computing and Archiving Research Environment (COARE) is a supercomputing facility managed by DOST-Advanced Science and Technology Institute. COARE provides large compute and cloud storage resources to the science community. These resources are provided through its three (3) main services, namely, High-performance Computing (HPC), Science Cloud, and Data Archiving. <https://coare.asti.dost.gov.ph/services>
- GeoRiskPH: <https://georisk.gov.ph/>
- PlanSmart Ready to Rebuild: <https://bit.ly/LaunchPlanSmart>
- Infrastructure Development Program (IDP): <https://pcieerd.dost.gov.ph/supported-programs-projects/strategic-thrusts-and-programs/240-institution-development>
- OneLab Program: <https://web.onelab.ph/>
- DOST OneExpert: <https://oneexpert.gov.ph/>
- Advanced Device and Materials Testing Laboratory (ADMATEL): <https://admatel.com/base/>
- Electronics Product Development Center (EPDC): <https://asti.dost.gov.ph/projects/epdc/>
- Advanced Manufacturing Center (AMCen): amcen.dost.gov.ph
- DOST-PCAARRD Advanced Learning Management System (PALMS) – An online learning platform offering training materials, webinars, and capacity-building courses to equip researchers, extension workers, and other stakeholders with updated knowledge and skills in AANR research and innovation: <https://www.pcaarrd.dost.gov.ph/index.php/quick-information-dispatch-qid-articles/advancing-educational-frontiers-palms-asynchronous-courses-launch-at-the-national-science-technology-and-innovation-week-2023>

48. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

☐ Yes

☒ No

49. 3.5. cont. **If no**, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

☐

Yes

☐

No

☒

This needs further review and assessment

50. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

☒

Yes

☐

No

51. 3.6 cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

• The National Copernicus Capacity Support Action Programme for the Philippines (CopPhil) aims to deploy a local infrastructure dedicated to the efficient distribution of large datasets in a controlled cloud environment. Through an effective combination of open-source tools and a software optimized for discovering, cataloguing, and using Earth Observation data, the CopPhil Infrastructure will provide free, open, and immediately accessible information services through a Copernicus Data Centre. - <https://copphil.philsa.gov.ph/copphil-infrastructure/>; https://asti.dost.gov.ph/resources/media-release/first-in-the-region-philippines-eu-join-forces-for-copphil-implementation-strengthens-disaster-response/?fbclid=IwZXh0bgNhZW0CMTEAAR0krf3tgUayK3NA6-kgIPWoNgawxspL2vSIOV1vSEmgV7fIM7i9plx8R5A_aem_9zqazC71gyLtrV-6VUQ • ASTNET (ASEAN Science and Technology Network). Info can be found here: <https://framestride.com/astnet/about-us/> • ASEAN Network for Drugs, Diagnostics, Vaccines and Traditional Medicines Innovation (ASEAN-NDI): <http://www.asean-ndi.net/> • ASEAN Journal on Science and Technology for Development (AJSTD): <https://ajstd.ubd.edu.bn/> • ASEAN Talent Mobility: <https://asean-talent.nxpo.or.th/home> • The East Asia Science and Innovation Area Joint Research Program, or e-ASIA JRP, is a multilateral international joint initiative between a number of public funding organizations of the East Asia Summit (EAS) member countries*. * EAS members include: 10 ASEAN member countries and 8 additional countries (Australia, Japan, New Zealand, China, India, South Korea, Russia, U.S.) (EAS participating countries) eASIA Projects can be found here: <https://www.the-easia.org/jrp/projects.html> • The Medipix technology is one of CERN's most successful knowledge transfer cases. Each collaboration has triggered a significant number of commercial activities in widely differing application areas. Medical imaging, space dosimetry, education, and material analysis just to mention a few. Moreover, the industrial partners and license holders commercialising the Medipix technology range from established enterprises to start-up companies. Thus, the technology is contributing to value creation, not only in its application areas, but also through business creation. <https://medipix.web.cern.ch/about-0>

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

52. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

☒ Yes

☐ No

53. 4.1 cont. **If yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

• Institution Development Program (IDP) aims to strengthen the research capabilities of academic and research institutions by supporting the development and upgrading of research laboratories and facilities. • Regional Research Institution (RRI) encourages new researchers and institutions to engage in research projects that address regional needs. • Good Governance through Data Science and Decision Support Systems (GODDESS) Program assists in developing systems and technologies to support data-driven governance and evidence-based management. (Formal: <https://pcieerd.dost.gov.ph/supported-programs-pprojects/strategic-thrusts-and-programs/239-human-resource-development-2#xi-good-governance-through-data-science-and-decision-support-system-goddess>) • Balik Saliksik Program provides grants to new PhD graduates who specialize in areas aligned with DOST-PCIEERD's priorities. https://pcieerd.dost.gov.ph/images/2024/guidelines/SignedrevisedBalikSaliksikGrant2022_1-22-0428-40.pdf • The EXPERTISE (Expert Intervention for Scientific Engagement (ExpertiSE) Program Offers scientific skills and services of all our consortia researchers to the industry in the region supports budding and emerging industries and/or companies in the region and identify their potential niche, industry gaps, challenges and R&D needs as well as engage the industry leaders in each region. More about different capacity building initiatives from: <https://pcieerd.dost.gov.ph/supported-programs-pprojects/strategic-thrusts-and-programs/239-human-resource-development-2#coverage-2> • Government STEM undergraduate and graduate scholarship offerings: <https://science-scholarships.ph/>

54. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

☐ Yes

☒ No

55. 4.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

The values and principles of open science have long been embedded in the national frameworks and policies. However, the country has yet to further promote the UNESCO Recommendation on Open Science and how this can specifically impact the current policies and strategies of the country.

56. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

☐ Yes

☒ No

57. 4.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

There is limited time to consult other ministries/national government agencies on this aspect.

58. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

- ☐ Yes
- ☐ Partly
- ☒ No

59. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

- ☒ Yes
- ☐ No

60. 4.5 cont. **If yes**, please provide more information and links as relevant.

• Science Communication for Innovation Program: https://pcieerd.dost.gov.ph/images/2024/roadmap/IG/Science_Communication_for_Innovation_Roadmap_v3.pdf • Joint Harmonized Science Communication initiatives of the DOST and the University of the Philippines Los Baños: <https://uplb.edu.ph/all-news/cdc-presents-harmonized-science-communication-framework-for-filipinos/> • Science and Technology Academic and Research-Based Openly Operated KioskS (STARBOOKS): <https://www.starbooks.ph/> • Bantog Awards - The Science For The People Media Awards: <https://www.dost.gov.ph/knowledge-resources/news/84-2024-news/3639-bantog-awards-for-media-and-science-communication-general-guidelines.html> • DOST, through its attached agencies, conducts programs/activities on research dissemination, utilization and policy translation. In addition, the National Academy of Science and Technology (NAST) is an advisory body to the President of the Republic of the Philippines on policies concerning science and technology in the country, and the National Research Council of the Philippines (NRCP) provides scientific advice on problems and issues of national interest.

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

61. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (*ref.: (v) 20.b, 20c, 20i*)

☐ Yes

☒ No

62. 5.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

There is limited time to consult other universities/agencies on this aspect.

63. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

☐ Yes

☒ No

64. 5.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

There is limited time to consult other ministries/agencies on this aspect.

65. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

☒ Yes

☐ No

66. 5.3 cont. **If yes**, for each initiative, please provide more information, including on the stakeholders[i] that are involved or lead the initiative, and links as relevant.

[i] For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.

This might not be strictly limited to "open science" per se, but to the practices that promote open science from the National Academy of Science and Technology (NAST) of the Philippines: (i) Outstanding Scientific Paper Awards: <https://recognition.nast.ph/index.php/nast-awards/outstanding-scientific-paper-awards>; (ii) DOST International Publications Award: <https://recognition.nast.ph/index.php/dost-awards/dost-international-publications-award> - The DOST Intellectual Property Award - International Publication Award (DOST IPA-IPub) is given annually to scientists and researchers of the Department of Science and Technology (DOST) who were able to publish their research outputs in Web of Science/Clarivate Analytics or Scopus indexed journals in the last three (3) years before the year of the nomination.; (iii) Magsaysay Future Engineers/Technologists Award: <https://recognition.nast.ph/index.php/dost-awards/magsaysay-future-engineers-technologists-award> - The Magsaysay Future Engineers/Technologists Award aims to encourage young Filipino students to pursue a career in science, engineering, and technology and to recognize outstanding research outputs on engineering and technology at the collegiate level. (iv) The Outstanding Young Scientist (OYS) Awards are given to young Filipino scientists (must not be 41 years old within January to December in the year of the awarding) who have made significant contributions to science and technology. <https://recognition.nast.ph/index.php/nast-awards/outstanding-young-scientist>; (v) NAST Talent Search for Young Scientists: <https://recognition.nast.ph/index.php/nast-awards/nast-talent-search-for-young-scientists> - The NAST Talent Search for Young Scientists is a project of NAST to encourage young people to pursue a career in science. Trophies and cash prizes will be awarded to the best three scientific paper presenters. The paper may present basic or applied research.

67. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

☐ Yes

☒ No

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES

OF THE SCIENTIFIC PROCESS

68. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21)

☒ YES, at the national level

☐ YES, at the institutional level

☐ YES, at both the national and institutional levels

☐ No

69. 6.1 cont. **If yes**, please indicate which of the recommended actions they address:

- ☐ promotion of pre-prints
- ☐ exploration of open peer-review practices
- ☐ valuing and sharing of negative scientific results and associated data
- ☐ participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- ☒ development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas
- ☐ promoting open innovation practices
- ☐ Other

70. 6.1. cont. **If yes**, please provide more information and links as relevant

The development and updating of the Harmonized National Research and Development Agenda (HNRDA) are spearheaded by the agencies of the Department of Science and Technology, in consultation with government and private research institutions, the academe, industry, and other concerned agencies and stakeholders of respective sectors.

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

71. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

The Philippines, through the Department of Science and Technology (DOST) pursues science-based international collaborations across different fields of science, technology, and innovation. These partnerships are forged and strengthened through implementation of joint initiatives such as joint research and development projects, scientific visits/missions, and joint scholarship programs. These include collaborations with countries like Japan, United Kingdom, South Korea, India, China, and e-Asia, among others. While there are counterpart funding for the collaborations, there is no cross-border mechanism (transfer of funds). The most common area of cooperation for joint research projects are health, agriculture, and geology. The DOST has also partnered with Elsevier to train researchers on research publishing, promoting open science practices.

72. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

☐ Yes

☒ No

73. 7.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

This remains an area for future progress in the Philippines.

74. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

☐ Yes

☒ No

75. 7.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

There is limited time to consult other ministries/agencies on this aspect.

Part 8. GENERAL CONSIDERATIONS

76. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

☒ Yes

☐ No

77. 8.1 cont. **If yes**, please provide more information.

Yes, these include data and sample sharing concerns or limitations of other countries and expensive publication fees in journals

78. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org